

Rohan Kapur

rohanrajkapur@gmail.com | 310-849-3377 | [linkedin.com/in/rohan--kapur/](https://www.linkedin.com/in/rohan--kapur/) | rohankapur.github.io/portfolio/

EDUCATION

University of California Berkeley, School of Information

August 2024 – May 2026

- M.S. of Information and Data Science
- Relevant Coursework: Data Engineering, Applied Machine Learning, Machine Learning at Scale, Computer Vision, Causal Inference

Carnegie Mellon University, Tepper School of Business

August 2020 – May 2024

- B.S. in Business Administration (Analytics and Data Mining) and Decision Science

EXPERIENCE

University of California, Los Angeles (UCLA) Office of Research Information Systems

Data Architect

Jan 2026 – Present

- Built and deployed an ETL pipeline (SSIS, PowerShell, T-SQL) to capture critical Azure Extended Event sessions (deadlocks, ad hoc queries, long-running queries, and error traps) into a production data warehouse, replacing a manual weekly review that covered an estimated <1% of 100,000+ log entries; developing real-time alerting to reduce detection **time from 144 hours to 6 hours**.
- Reduced average runtime by 13% for 5 legacy SQL Jobs executing twice a day, cutting Azure SQL compute **costs by 1-2%**.
- Eliminated a weekly 4 hour manual task by automating publishing data warehouse updates using PowerShell and Power Automate.

Data Scientist

March 2024 – December 2025

- Led end-to-end redevelopment of Human Research Protection Program's analytical protocol reports, productionizing 37 of 45 reports. 5 of these reports were made into dynamic dashboards and distributed to customers, increasing user engagement by 15%.
- Initiated collaboration with internal Data Architecture and Data Analytics teams to streamline data ingestion and data interpretability, reducing SQL complexity by 3% across 70 queries and improving cross-team data reliability.
- Optimized Power BI audit report, improving data accuracy for 4 government regulated agencies and 3 C-Suite leaders. Enabled two-day audit turnarounds and ensured future audits are seamless through consistent data structures and visualizations.

Wells Fargo, Advanced Data Scientist Intern (Master's level program completed as undergraduate)

June 2023 – August 2023

- Contributed to a 12% increase in efficiency and 70% reduction in fraud rates for the PAZE marketing campaign. These contributions led a 20% increase in customer engagement by filtering 10TB of customer interaction and fraud claim data across multiple databases.
- Developed guides for 45-person team to assess the performance of causal inference packages: Causal & Econ ML and DoWhy, quantitatively comparing accuracy, computational efficiency, and accessibility on 15 customer datasets reducing processing by 25%.
- Pioneered team presentation with 5 ADAP colleagues on causal inference tools and initiated consolidation methods to remove redundant information while maintaining 100% of the key content, improving audience engagement by 30% with key stakeholders.

New York Life, Data Analytics Intern

June 2022 – August 2022

- Published Tableau Dashboard showing customers conversion trends from term life policy plans to whole life plans, showcasing a 20% average decline in whole life plan value every 10 months. Presented findings to the VP of Digital Experience and top agents.

KEY PROJECTS & ACCOMPLISHMENTS

Immigration Navigator - RAG-Powered Visa Guidance Chatbot

- Collaborated on a 4-person team to design a five-stage RAG pipeline with synonym expansion to bridge colloquial and legal terminology (e.g. "work permit" → "employment authorization document"), improving retrieval accuracy on immigration queries
- Indexed 3,000+ USCIS documents into a ChromaDB vector store, enabling personalized guidance across the F-1 → H-1B pipeline
- Outperformed Claude and ChatGPT by 11-24% on core immigration pipeline, F-1 status, H1B, and STEM OPT questions
- Architected AWS infrastructure (EC2, S3, IAM, KMS, Secrets), with FastAPI/Streamlit services on Nginx with SSL for public access.

California Housing Price Predictions – Caching and Kubernetes

- Built and containerized a FastAPI microservice that serves a scikit learn regression model predicting California housing prices from data wrangled inputs like median income, house age, latitude, and longitude, with single and bulk prediction endpoints.
- Implemented request caching in Redis and productionized on Kubernetes. Produced comprehensive tests with pytest and httpx, achieved 29 passing unit and integration tests, to validate the deployment with a Minikube cluster and Docker image.

Diabetic Retinopathy Detection – Machine Learning Neural Networks

- Engineered a convolutional neural network with TensorFlow to detect diabetic retinopathy from images, including class rebalance, image augmentation, early stopping and learning rate optimization. Improved validation accuracy to 84% with unseen data.

Landcover Classification – Machine Learning Decision Forests

- Devised an image classification pipeline using satellite data to distinguish land-cover types. Applied image augmentation and tuned random forest classifiers resulting in validation accuracy to 63% while maintaining less than a 10% train-validation gap.

SKILLS, LANGUAGES, & INTERESTS

- Skills: RAG, Computer Vision, Machine Learning, Kubernetes, AWS, SQL, Powershell, R, PowerBI, Big Data, Causal Inference
- Interests: Fashion (DIY projects), Sports (Basketball, Tennis, Yoga, Sports Analytics), Cryptocurrency/Web3, Digital Marketing